Hydroformylation of C_2H_4 over Rh-based SILP catalysts: molecular interactions and speciation studied by in situ IRAS and DFT

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Introduction

In recent years, ionic liquids (ILs) have been widely used as solvents for many chemical reactions due to their unique physicochemical properties.[1,2] The concept of supported ionic liquid phase (SILP) catalysts exploits ILs to dissolve a homogeneous transition metal catalyst and immobilize it as a supported thin film coating on a porous carrier.[3] Thus, SILP catalysts combine the advantages of both homogeneous and heterogeneous catalysis and have been explored for many industrially relevant reactions in recent years.[4] For instance, SILP systems with Rh as the active metal center (e.g. Wilkinson's catalyst) have been successfully applied in hydrogenation and hydroformylation reactions.[5] Here, bidentate phosphine ligands are crucial, as they stabilize the dissolved Rh complex against degradation such as in-situ formation of metal NPs.[6] Furthermore, the presence of the surrounding IL phase is beneficial in this context. In the present study, we focus on the molecular interactions in the SILP and elucidate the speciation of the molecular complex in the as-prepared catalyst and under hydroformylation conditions.

Materials and Methods

We choose the hydroformylation of C_2H_4 as a model reaction which yields a clear product profile. The applied SILP catalyst consists of [Rh(acac)$(CO)_2$] as catalyst precursor, a four-fold excess of 4,5-Bis-(diphenylphosphino)-9,9-dimethylxanthen (xantphos) as stabilizing ligand and [C_4C_1Im][NTf_2]. We follow a combined

experimental and computational approach to systematically investigate the SILP system and its speciation after preparation and under reaction conditions. In the first step, we evaluated the performance of the catalyst systems in terms of activity and selectivity in a fixed-bed reactor. By using dedicated IR-spectroscopic methods in combination with density functional theory (DFT), we obtain fundamental insights into the underlying processes. In particular, we apply transmission infrared spectroscopy (TIR) to elucidate the molecular interactions between individual constituents in the bulk IL phase. Furthermore, infrared reflection absorption spectroscopy (IRAS) was performed on model SILP systems in planar configuration to study thermal and reactant-induced effects. The support of the model SILP systems consists of a Si wafer customized with layers of Ti (2 nm), Au (100 nm), and SiO_2 beads (7-22 nm in diameter). These wafers ensure excellent wetting of the IL phase and high stability in terms of spectral quality during exposure to reactants or thermal treatment.

Results and Discussion

In the bulk IL phase, we observe the expected xantphos/acac ligand exchange but also the formation of $[NTf]_2^-$ stabilized Rh complexes. Thus, we expect the same processes during the preparation of the porous SILP material. Next, we mimicked the reaction conditions of the fixed-bed reactor experiments in our in-situ IRAS experiments and systematically investigated the effect of the xantphos ligand, H_2/CO, and H_2/CO/C_2H_4 dosing on the speciation in the SILP at different temperatures. We identify the Rh complexes as a function of the composition of the SILP and the reaction conditions. Our results provide insights into an extended network of Rh species (including mixed $[NTf_2]^-$ / xantphos / acac hydride and carbonyl complexes) which can be interconverted dynamically. For instance, application of reactive atmosphere (H_2/CO) leads to the formation of several hydride and carbonyl complexes, in particular $[HRh(CO)_2(xantphos)]$, the immediate precursor that enters the catalytic cycle upon loss of a CO ligand. The extension of the spectroscopic studies of SILP systems to (PM-)IRAS opens exciting possibilities for further studies.

References

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